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Integrated Processing of Borehole and Surface Seismic Measurements for Enhanced Well Tie and AVO Analysis

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SUMMARY

The Wisting discovery in the southwest Barents Sea is characterised by a shallow reservoir lying at as little as 250 m below the sea bed. At the time of this study three exploration wells had been drilled in the field. At the first two wells achieving a well tie between borehole and surface seismic measurements had proved challenging. At the latest well, the operators OMV acquired an extensive suite of borehole measurements including multi-azimuth Walk-Away VSP (WAVSP) and wireline sonic data to facilitate the well tie. Historically, borehole seismic and wireline acquired sonic data are processed separately to one another and to the surface seismic data with which they must ultimately tie. OMV chose to instead follow an integrated processing methodology with continuous calibration and validation of results across the different domains. This workflow results in consistent datasets which have been used to derive a 1D VTI elastic model of the subsurface at the well which ties all data across each domain. By processing both surface seismic and borehole data together in this manner a well tie has been achieved and a deeper understanding of key factors that influence well tie at the Wisting discovery has been gained.

Introduction

The Wisting discovery, operated by OMV, is a shallow reservoir some 250 m below the seabed, situated in the southwest Barents Sea. Downhole data from the first two exploration wells proved very difficult to tie with the existing 3D and 2D surface seismic data, which hampered extrapolating information from the wells to the wider Wisting area and impacted further reservoir characterization. When drilling the third exploration well, OMV acquired an extensive suite of borehole measurements, including a multiazimuth walkaway vertical seismic profile (WA-VSP) and wireline sonic data. These were intended to provide detailed estimates of the anisotropic elastic properties of the overburden and reservoir to assist with understanding the well tie. Conventionally, borehole seismic measurements, wireline data, and surface seismic data are processed independently with little attempt to ensure consistency between data sets. OMV chose instead to maximize consistency by processing all available data from the well location in an integrated manner with continuous cross-domain calibration and validation of results.

Integrating borehole and surface seismic data

Our workflow uses all available data to generate, at the well location, a fully elastic 1D model of the subsurface from surface to terminal datum. Elastic finite-difference synthetic modelling is used to predict the surface and borehole seismic responses from the model. Discrepancies between the modelled responses and the measurements indicate deficiencies that are addressed by editing and reprocessing the relevant data sets. The process proceeds iteratively until the predictions match the measurements, or until the remaining discrepancies are explained. This model-based approach, with input from a variety of disciplines, enables integration of data sets of disparate types, scales, bandwidths, and geometries.

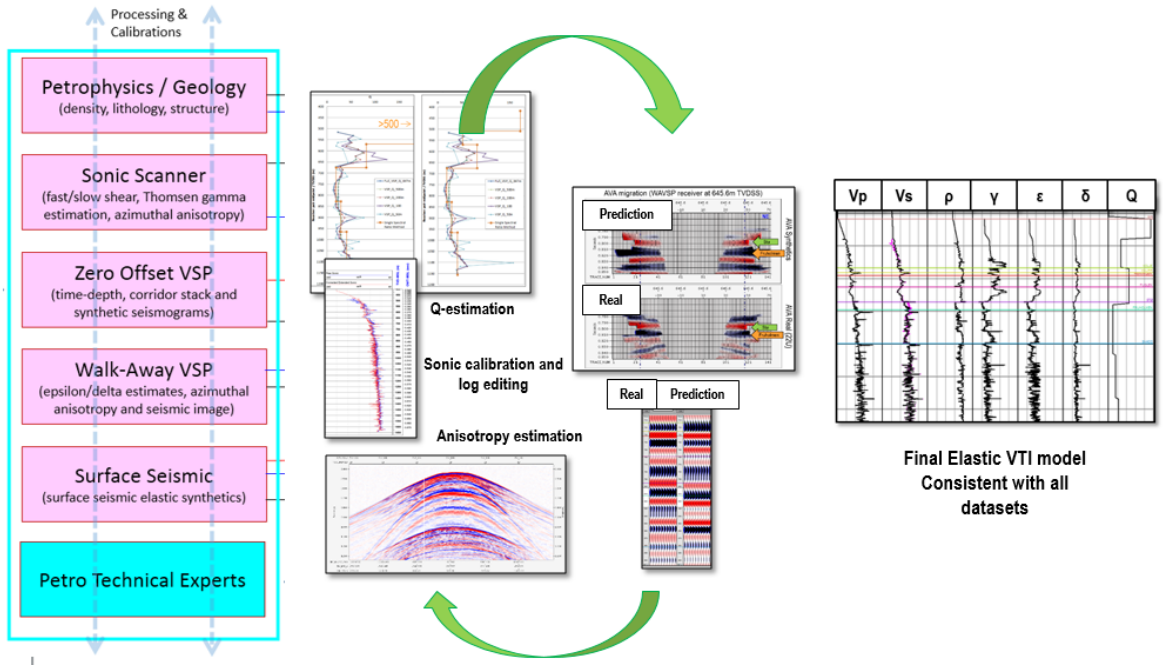


Figure 1 *A schematic diagram of the integrated workflow, showing how all available information is combined to create a 1D elastic model consistent with all the measurements.*

Applying the workflow to the Wisting discovery

OMV's third exploration well, named Hanssen, was drilled in mid-2014, and a comprehensive suite of downhole measurements was acquired. This included wireline measurements, a zero-offset vertical seismic profile (ZO-VSP) and a multiazimuth walkaway VSP (WA-VSP). These complemented the existing surface seismic data that comprised a 3D field-wide towed-streamer seismic survey and two

orthogonal high-resolution 2D towed-streamer lines from the pre-drill site survey. Table 1 shows how each data set contributed to the elastic 1D model.

Elastic Property	Source	Calibration	Validation
V_p	Wireline sonic compressional	Zero offset VSP	Synthetic modelling and surface seismic
V_s	Wireline sonic shear	Rock Physics (by OMV)	Synthetic modelling and AVO response of surface seismic and WAVSP
Density	Wireline density		Synthetic modelling and surface seismic
Thomsen's Epsilon	WAVSP	Sonic derived Gamma and surface seismic	Synthetic modelling, surface seismic and WAVSP
Thomsen's Delta	WAVSP	Sonic derived Gamma and surface seismic	Synthetic modelling, surface seismic and WAVSP
Thomsen's Gamma	Wireline sonic	WAVSP and surface seismic	Synthetic modelling and AVO response of surface seismic and WAVSP
Q_p	Zero offset VSP	Surface seismic	Synthetic modelling and AVO response of surface seismic and WAVSP

Table 1 *A summary of the information provided by the data sets input to the integrated workflow.*

Model creation proceeded in a step-wise iterative fashion. It was first populated, at 0.5-m depth sampling, with velocity and density information from the wireline measurements. The velocity data from the sonic were then calibrated to seismic bandwidth using the time-depth curve from the ZO-VSP. The ZO-VSP was processed to produce a corridor stack that was compared with the reflectivity series from wireline data. This was then validated by comparison with the high-resolution 2D seismic lines, as available reflection angles from the 3D seismic data covered only angles beyond 20°, violating the zero-offset assumption. Any adjustments required to achieve consistency were investigated, cross-domain validated, and then applied to the model and VSP processing flow.

Next, the epsilon and delta curves were added by means of a slowness-polarization inversion of the WA-VSP (Leaney & Hornby, 2007). However, the shallow target and the 200-m exclusion zone around the drilling platform, in which the source vessel could not operate, meant that the available reflection angle range at target depth was limited to about 30° to 60°. This lack of near angles hampered the inversion, but we were able to stabilize it using sonic-derived estimates of Vs and Thomsen's gamma. The WA-VSP provided six discrete anisotropy estimates between depths of 640 and 860 m, but the sonic data provided a continuous high-resolution gamma estimate (Walsh *et al.*, 2007) over a much larger depth range. By calibrating the gamma curve with the WA-VSP-derived epsilon and delta estimates, we could generate epsilon and delta curves over that depth range. This proved especially important in the shallow overburden where very high anisotropy was indicated, and this was confirmed by forward modeling of the surface seismic data and WA-VSP. Isotropic and anisotropic depth imaging of the surface seismic data were used from an early stage to validate the anisotropy updates. The ZO-VSP provided high-resolution estimates of anelastic absorption ("Q") and suggested that substantial absorption was occurring in the shallow overburden. This was validated by including amplitude and phase effects of absorption during depth imaging of the surface seismic data.

Results

By requiring that discrepancies be explained, the integrated workflow ensures that each data set is analyzed in more detail than when they are processed in isolation. The primary success criterion of this study was that a tie be achieved between the surface seismic and borehole measurements at the Hanssen well. Figure 2 shows that this objective was largely achieved in terms of event timing and AVO/AVA response. Several factors contributed to this:

- *Elastic VTI anisotropy and Q*: Only by including the overburden VTI anisotropy and Q (Cavalca *et al.*, 2013) in depth imaging were we able to get an accurate tie to the synthetic AVO response (Figure 3). Simultaneous integration of multiple information sources was key to this success
- *WA-VSP processing*: The borehole seismic and surface seismic data were processed in the same seismic software package, and knowledge-sharing between borehole and surface seismic processors enabled attenuation of acquisition-related noise from the WA-VSP data at important depth levels, as well as helping ensure consistency between data sets.

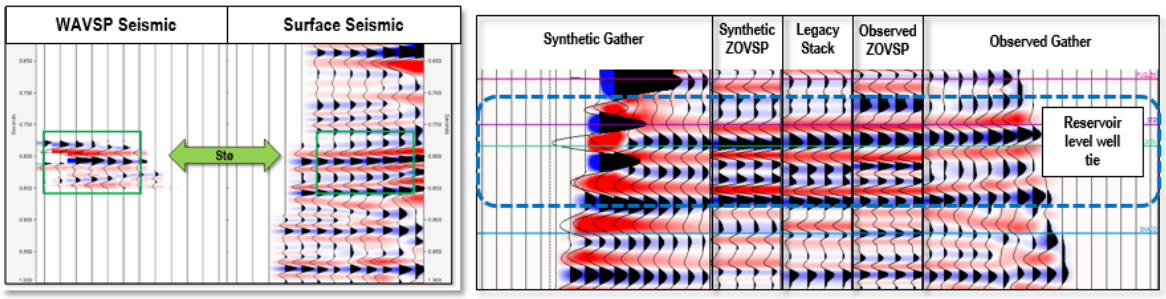


Figure 2 *An illustration of the consistency of the data sets resulting from this processing (left) and the final well tie between the synthetic and observed surface seismic and ZOVSP data.*

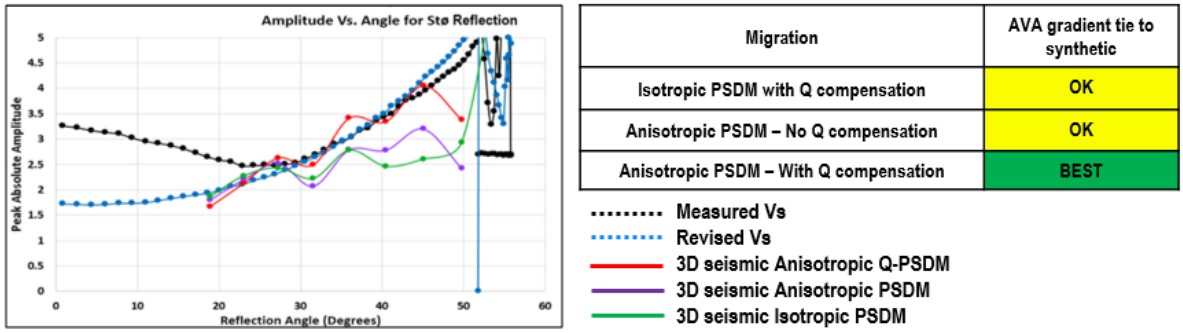


Figure 3 *A comparison of the reservoir AVO post imaging vs. AVO from elastic synthetics. The two synthetic profiles use two different shear velocities. Anisotropic prestack depth migration (PSDM) including Q compensation results in the best tie with the synthetic data.*

- *Interbed multiple identification and analysis*: The shallow Wisting reservoir occurs above the first water layer multiple in the surface seismic data, and so the impact of multiples on AVO analysis had not previously been considered. Full-waveform modeling of the surface seismic gathers clearly showed that the reservoir reflections were contaminated by interbed multiples originating in the overburden, and this was confirmed by careful inspection of the surface seismic gathers. Identifying the issue enabled us to propose mitigation procedures for future processing.
- *Vs modification within the reservoir*: Synthetic modelling showed that using a field-wide rock physics-based reservoir Vs better matched the reservoir AVA response. The limited near-angle coverage of the surface seismic data gave an uncertain AVA intercept at the reservoir and it proved essential to model additional reflections to calibrate the AVA response (Figure 4). We continue to investigate this discrepancy in the measured vs. modelled Vs. One hypothesis is that that the sonic- and seismic-scale measurements are 'seeing' the formation Vs differently.

Having an accurate VTI model at the Hanssen well location enabled extracting significantly more information from the surface seismic data. Figure 5 shows the intersection between the two high-resolution 2D lines after anisotropic Q prestack depth migration. A clear oil/water contact can now be seen, which had not been observed previously.

Conclusions

The integrated processing of borehole seismic data, wireline sonic log data, and surface seismic data at the Hanssen well enabled us to cross-calibrate and validate results from the different domains. This led to the accurate determination of elastic VTI properties, and of Q, that have significantly improved the prestack AVO/AVA well tie, improved the zero-offset well tie and led to improved seismic images. The origin and impact of interbed multiples on the AVO response has been demonstrated via the integrated worfklow. These results contributed to greater understanding the Wisting area as the discovery moves into further development.

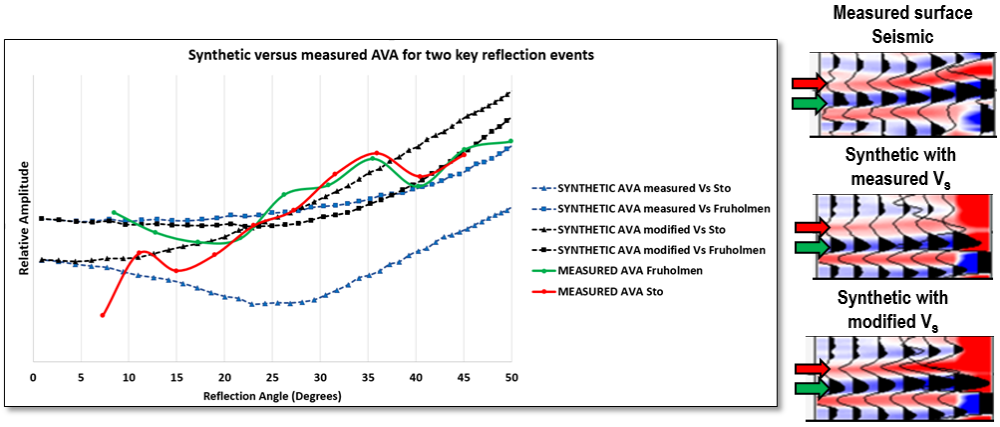


Figure 4 *An illustration of the measured versus predicted Stø and Fruholmen reflection AVA. The predicted AVA results show the differences introduced by using the modified Vs in place of the measured Vs. All other elastic properties remain unchanged.*

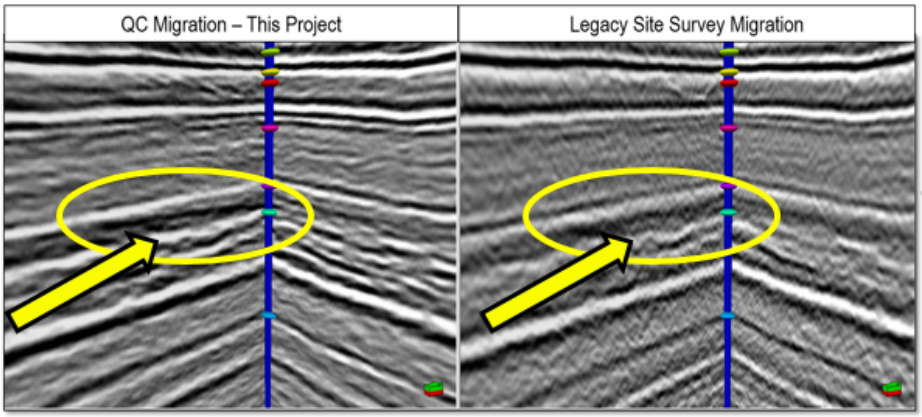


Figure 5 *A comparison of the legacy time site survey image (right) with the Q-PSDM image from this study (left). The yellow circle highlights the location of the oil/water contact within the reservoir that is revealed on the left image.*

Acknowledgements

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